

Azure & Security Progress — Sept 17, 2026

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Overview

Two threads of work today, both cloud security training.

First, built **The Jekyll & Hyde Files** — a persisted, dual-role Azure Administrator + Cloud Security Analyst training simulator, covering 10 hands-on labs per role plus 5 combined build-then-investigate scenarios.

Second, hands-on infrastructure and detection work in **The Ward**, a personal Microsoft Sentinel lab: stood up the workspace, connected Azure control-plane and Entra ID identity telemetry, ran first KQL investigations, and shipped the first analytics rule — moving from ad hoc portal clicking to an actual working telemetry-to-detection pipeline.

Built today: The Jekyll & Hyde Files

A training simulator built around a dual identity: Dr. Jekyll (Azure Administrator, builds and operates the environment) and Mr. Hyde (Cloud Security Analyst, investigates it without assuming what happened). A "Hand off to Hyde" action plays a deliberate transition and locks in the psychological separation between roles — you did not build what you're now investigating.

What's inside:

- **10 Jekyll labs** — resource groups, Entra identity, RBAC, VNets/NSGs, VM hardening, storage, App Service, Azure Monitor, governance
- **10 Hyde labs** — Defender for Cloud posture, VM and storage security, identity and RBAC investigation, network investigation, vulnerability triage, Sentinel/KQL investigation, incident response
- **5 combined Jekyll → Hyde scenarios** — build a resource, then hand off and investigate what actually happened to it, with the Hyde half gated until handoff
- **Progressive KQL exercises**, PowerShell/CLI challenges, 49 vocabulary terms with flashcards, two skill trees, an AI-graded Interview Mode and "Can I Explain It?" mode, and an incident report template that exports

- A **16-track progress dashboard** that shows what's actually been practiced rather than a score

Every scenario question hides its answer (reasoning, evidence, investigation path, remediation) behind a click, and nothing in the app can be checked off without doing the underlying work.

[Open the app](#) · progress saves locally in the browser. A standalone offline copy was also packaged for local use.

The Ward — Azure & Sentinel lab progress

Hands-on infrastructure and detection work in The Ward, a personal Microsoft Sentinel lab. Worked primarily from the Microsoft Defender portal rather than the older Azure-portal Sentinel workflow, and got noticeably more comfortable navigating Defender, Sentinel, Content Hub, Data Connectors, Hunting/Advanced Hunting, Analytics, the Log Analytics workspace, and Entra ID.

Infrastructure

- Created the `rg-the-ward` resource group for the security lab
- Created the `law-the-ward` Log Analytics workspace
- Enabled Microsoft Sentinel on the workspace

Azure Activity — control-plane telemetry

- Installed the Azure Activity Sentinel solution from Content Hub
- Configured the Azure Activity data connector and connected the subscription to `law-the-ward`
- Generated a deliberate control-plane event (modified a resource tag) and confirmed it landed in `AzureActivity`
- Queried Azure Activity successfully from Defender Advanced Hunting, and learned a real schema distinction between Advanced Hunting and traditional Log Analytics/Sentinel queries — inspect the table's schema (`getschema`) instead of assuming column names

```
flowchart TD
```

```
A[Azure Resource] --> B[Azure Activity]
```

```
B --> C[Sentinel]
```

```
C --> D[Log Analytics]
```

```
D --> E[AzureActivity table]
```

```
E --> F[KQL]
```

Microsoft Entra ID — identity telemetry

- Worked with test users and security groups; confirmed the Entra ID data connector is connected and Audit Logs are enabled for the Sentinel workspace
- Verified the `AuditLogs` table exists in `law-the-ward` — found it empty at first, then generated fresh Entra activity and confirmed ingestion
- Inspected a real event (`groups0DataV4_Get`) and identified the fields that matter for investigation: `TimeGenerated`, `OperationName`, `Result`, `InitiatedBy`, `TargetResources`, `AdditionalDetails`

```
flowchart TD
```

```
A[Entra ID] --> B[Audit Logs]
```

```
B --> C[Microsoft Sentinel]
```

```
C --> D[law-the-ward]
```

```
D --> E[AuditLogs table]
```

```
E --> F[KQL]
```

KQL / log investigation

Ran real KQL, not just read about it:

```
AuditLogs
```

```
| where TimeGenerated > ago(30m)
```

```
| order by TimeGenerated desc
```

```
AzureActivity
```

```
| where TimeGenerated > ago(24h)
```

```
| order by TimeGenerated desc
```

Also learned: `TimeGenerated` is the relevant timestamp on both tables; Advanced Hunting doesn't necessarily expose Sentinel tables under the same names; `AuditLogs` is a Log Analytics/Sentinel table while `ASimAuditEventLogs` is an ASIM-normalized table and not an automatic substitute for it; `EntraIdSignInEvents` and `IdentityInfo` weren't

available in the current Advanced Hunting environment — a reminder not to assume the docs' schema matches the tenant's actual schema. The habit that mattered most: inspect the data you actually have before writing the query you wish you had.

Sentinel detection engineering

- Created the first Sentinel analytics rule for The Ward, configured as a scheduled query rule
- Got the KQL/rule configuration working after troubleshooting the editor

```
flowchart LR
  A[Telemetry] --> B[KQL]
  B --> C[Analytics Rule]
  C --> D[Alert]
  D --> E[Incident]
```

This is the point The Ward stopped being just a logging lab and started becoming a SIEM lab.

Azure concepts reinforced

Deliberately built the AZ-104 mental map instead of just clicking through the portal — reviewed resource groups, subscriptions, Entra ID (tenants, users, groups, RBAC, authentication vs. authorization), Azure Monitor (activity log, metrics, alerts, workbooks), Log Analytics, Sentinel, Defender for Cloud, storage (blob/files/queue/table), networking (VNETs, subnets, NSGs, private endpoints, public IPs, load balancers, App Gateway, DNS, private DNS), compute (VMs, scale sets, App Service, Functions, containers, AKS), Logic Apps, Azure Automation, Key Vault, managed identities, Azure Policy, resource locks, tags, cost management, Advisor, ARM, Bicep, CLI, PowerShell, Cloud Shell, Bastion, Network Watcher, Azure Firewall, VPN Gateway, ExpressRoute, Front Door, and the IaaS/PaaS/SaaS distinction. Portal navigation is visibly getting faster — enough to deliberately pause hands-on work and turn the growing service list into flashcards.

Biggest accomplishment today

Went from clicking around Sentinel trying to figure out what things do, to a working pipeline:

```

flowchart TD
    W[THE WARD]
    W --> AA[Azure Activity<br/>control plane]
    W --> EI[Entra ID<br/>identity plane]
    AA --> AAT[AzureActivity]
    EI --> AL[AuditLogs]
    AAT --> LAW[law-the-ward]
    AL --> LAW
    LAW --> SEN[Microsoft Sentinel]
    SEN --> KQL[KQL]
    KQL --> AR[Analytics Rule]
    AR --> INC[Alert / Incident]
    INC --> AUTO[Automation + Playbook]

```

Completed tonight

- ☒ ~~Resource group, Log Analytics workspace, Sentinel enabled~~
- ☒ ~~Azure Activity solution + connector installed, test event generated and verified in AzureActivity~~
- ☒ ~~Entra ID connector confirmed, Audit Logs enabled, Entra activity generated and verified in AuditLogs~~
- ☒ ~~First analytics rule created and configured as a scheduled query rule~~

Next for The Ward

- ☐ Generate a deliberate event for the analytics rule
- ☐ Confirm the resulting Sentinel alert/incident
- ☐ Investigate the incident
- ☐ Build first automation rule
- ☐ Build first Logic App / playbook
- ☐ Connect automation rule → playbook
- ☐ Test automated response
- ☐ Add controlled remediation
- ☐ Add storage / data-plane telemetry
- ☐ Build additional detections
- ☐ Document the complete Ward architecture

Next session can legitimately start with incident investigation and SOAR — a good stopping point for today.